

SEFI Space: A Geometric Identity Space for Single-Entity Field Theory

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Abstract

The Single Entity Field Interpretation (SEFI) extends geometric worldline approaches to quantum field theory by introducing a structured identity space associated with a single underlying entity. In this work we define SEFI Space (SS) as a four-dimensional geometric identity space spanned by basis directions corresponding to FIELD ORIGIN, FIELD AUTHORSHIP, FIELD SOVEREIGNTY, and WARP EXPRESSION. The SEFI field is represented as a worldline-based construction embedded in spacetime, while its identity content is encoded as a trajectory in SEFI Space. We introduce a metric on SEFI Space, define identity velocity and acceleration, and formulate sovereignty and warp constraints as invariance and motion within this identity space. The resulting framework provides a geometric interpretation of field identity, stability, and transformation compatible with worldline quantum field theory and suggests a path toward unified single-entity field models.

1 Introduction

Worldline formulations of quantum field theory provide an alternative representation of particle and field dynamics in terms of path integrals over trajectories in spacetime [1, 2, 3]. These approaches have proven useful for perturbative calculations, effective actions, and the geometric interpretation of quantum amplitudes. However, the ontological status of field identity, superposition, and entity persistence remains largely external to the formalism.

The Single Entity Field Interpretation (SEFI) proposes that a given quantum field, such as the electron field, arises from a single continuous entity whose worldline structure spans all allowed trajectories in spacetime. The field is then understood as the projection of this worldline ensemble, with forward-time segments corresponding to particles and backward-time segments corresponding to antiparticles, in line with established worldline interpretations of electron–positron dynamics [2, 3].

In previous work, SEFI introduced layered identity constructs under the labels FIELD ORIGIN, FIELD AUTHORSHIP, FIELD SOVEREIGNTY, and WARP EXPRESSION. These layers were treated as conceptual components of the field’s identity structure. In this manuscript, we formalize these layers as basis directions of a geometric identity space, which we call SEFI Space (SS). SEFI Space is not a spacetime manifold, nor a gauge or spinor space; it is an abstract geometric space in which the identity of the single entity evolves as the worldline traverses spacetime.

Our goals are:

- To define SEFI Space as a four-dimensional real vector space associated with the SEFI field.
- To introduce a metric structure on SEFI Space and define identity velocity and acceleration.
- To express sovereignty and warp constraints as invariance and motion within SEFI Space.
- To couple SEFI Space to worldline geometry in a manner compatible with established worldline QFT.

2 Worldline Foundations

We briefly recall the worldline formulation relevant to SEFI. Let spacetime be a manifold M with metric $g_{\mu\nu}$. A relativistic point particle of mass m moving in M is described by a worldline $y^\mu(\tau)$ and an action of the form

$$S_{\text{WL}}[y] = -m \int d\tau \sqrt{-g_{\mu\nu}(y(\tau)) \dot{y}^\mu(\tau) \dot{y}^\nu(\tau)}, \quad (1)$$

where τ is an affine parameter along the worldline and dots denote derivatives with respect to τ [1].

Spin degrees of freedom can be incorporated using Grassmann variables $\psi^\mu(\tau)$, leading to a spin action of the form

$$S_{\text{spin}}[\psi] = \int d\tau \psi_\mu(\tau) \dot{\psi}^\mu(\tau), \quad (2)$$

and coupling to an electromagnetic potential $A_\mu(x)$ is introduced via

$$S_{\text{EM}}[y] = q \int d\tau \dot{y}^\mu(\tau) A_\mu(y(\tau)), \quad (3)$$

where q is the charge of the entity [2, 3].

The total worldline action for the entity is then

$$S_{\text{entity}}[y, \psi] = S_{\text{WL}}[y] + S_{\text{spin}}[\psi] + S_{\text{EM}}[y]. \quad (4)$$

In a path-integral formulation, the presence density of the entity at spacetime point x can be written schematically as

$$\rho(x) = \int \mathcal{D}y \mathcal{D}\psi e^{iS_{\text{entity}}[y, \psi]} \delta^{(4)}(x - y(\tau)), \quad (5)$$

and an emergent field amplitude may be expressed as

$$\phi(x) = \int d\tau \delta^{(4)}(x - y(\tau)). \quad (6)$$

In SEFI, the worldline is not merely a computational device but is taken to represent the actual geometric structure of a single entity whose projection yields the observed field. The SEFI Space introduced below provides a complementary description of the entity's identity content.

3 Definition of SEFI Space

We define SEFI Space (SS) as a four-dimensional real vector space associated with the SEFI field. The space is spanned by four basis directions corresponding to the SEFI identity layers:

$$\text{SS} = \text{span}\{\hat{O}, \hat{A}, \hat{S}, \hat{W}\}, \quad (7)$$

where

- $\hat{O}$ is the Origin direction, associated with FIELD ORIGIN,
- $\hat{A}$ is the Authorship direction, associated with FIELD AUTHORSHIP,
- $\hat{S}$ is the Sovereignty direction, associated with FIELD SOVEREIGNTY,
- $\hat{W}$ is the Warp direction, associated with WARP EXPRESSION.

An identity state of the entity at parameter value t along the worldline is represented by a vector $\vec{I}(t)$ in SEFI Space:

$$\vec{I}(t) = I_O(t)\hat{O} + I_A(t)\hat{A} + I_S(t)\hat{S} + I_W(t)\hat{W}, \quad (8)$$

where $(I_O(t), I_A(t), I_S(t), I_W(t)) \in \mathbb{R}^4$ are the identity coordinates at t .

The functions $I_O(t)$, $I_A(t)$, $I_S(t)$, and $I_W(t)$ are constructed from the worldline geometry and field layers. For example, one may define

$$I_O(t) = O(r(t)), \quad I_A(t) = A(r(t), \kappa(t)), \quad I_S(t) = S(r(t)), \quad I_W(t) = W(r(t)), \quad (9)$$

where $r(t)$ is a parametrization of the worldline, $\kappa(t)$ is its curvature, and O , A , S , and W are layer functions associated with FIELD ORIGIN, FIELD AUTHORSHIP, FIELD SOVEREIGNTY, and WARP EXPRESSION, respectively.

SEFI Space is not a spacetime manifold and does not carry a direct metric inherited from $g_{\mu\nu}$. Instead, we introduce an independent metric structure on SS to quantify changes in identity.

4 Metric Structure on SEFI Space

We equip SEFI Space with a diagonal metric of the form

$$ds_{\text{SS}}^2 = \alpha dI_O^2 + \beta dI_A^2 + \gamma dI_S^2 + \delta dI_W^2, \quad (10)$$

where $\alpha, \beta, \gamma, \delta > 0$ are constants that set the relative weight or stiffness of each identity layer. For an infinitesimal change in identity state

$$d\vec{I} = dI_O \hat{O} + dI_A \hat{A} + dI_S \hat{S} + dI_W \hat{W}, \quad (11)$$

the squared norm in SEFI Space is

$$\|d\vec{I}\|_{\text{SS}}^2 = \alpha dI_O^2 + \beta dI_A^2 + \gamma dI_S^2 + \delta dI_W^2. \quad (12)$$

Along the worldline, we define identity velocity and identity acceleration by

$$\dot{\vec{I}}(t) = \frac{d\vec{I}}{dt} = \dot{I}_O(t) \hat{O} + \dot{I}_A(t) \hat{A} + \dot{I}_S(t) \hat{S} + \dot{I}_W(t) \hat{W}, \quad (13)$$

$$\ddot{\vec{I}}(t) = \frac{d^2\vec{I}}{dt^2} = \ddot{I}_O(t) \hat{O} + \ddot{I}_A(t) \hat{A} + \ddot{I}_S(t) \hat{S} + \ddot{I}_W(t) \hat{W}. \quad (14)$$

The squared norm of the identity velocity is

$$\|\dot{\vec{I}}(t)\|_{\text{SS}}^2 = \alpha \dot{I}_O(t)^2 + \beta \dot{I}_A(t)^2 + \gamma \dot{I}_S(t)^2 + \delta \dot{I}_W(t)^2, \quad (15)$$

and similarly for the identity acceleration.

This metric structure allows us to define identity distances, identity curvature, and identity geodesics within SEFI Space, independent of the spacetime geometry.

5 Sovereignty and Invariance in SEFI Space

FIELD SOVEREIGNTY was originally introduced as a constraint ensuring that the field maintains its identity under allowed transformations. In the worldline formulation, one may express sovereignty as

$$F(T(r(t))) = F(r(t)), \quad (16)$$

for a class of transformations T that preserve the field identity.

In SEFI Space, sovereignty can be expressed as invariance of the identity vector under the induced action of T :

$$\vec{I}_T(t) = \vec{I}(t), \quad (17)$$

for all allowed transformations T . Here $\vec{I}_T(t)$ denotes the identity state obtained when the worldline is transformed by T and the layer functions are evaluated accordingly.

The set of transformations that preserve $\vec{I}(t)$ defines a symmetry group acting on spacetime and worldline geometry while leaving SEFI Space invariant. This group may be viewed as an identity-preserving subgroup of the full transformation group acting on the worldline. Sovereignty thus acquires a geometric interpretation as invariance in SEFI Space.

6 Warp as Motion in SEFI Space

WARP EXPRESSION was introduced as the layer responsible for geometric transformation of the field while preserving its identity. In the worldline picture, a warp operation W acts on the worldline as

$$r'(t) = W(r(t)), \quad (18)$$

modifying the geometry of the trajectory.

In SEFI Space, warp corresponds to motion along the Warp direction. If the warp layer function changes from $I_W(t)$ to $I'_W(t) = I_W(t) + \Delta I_W(t)$ under a warp operation, then the identity state becomes

$$\vec{I}'(t) = I_O(t)\hat{O} + I_A(t)\hat{A} + I_S(t)\hat{S} + I'_W(t)\hat{W}. \quad (19)$$

Warp is thus represented as a displacement in SEFI Space along $\hat{W}$, constrained by sovereignty requirements that ensure the overall identity of the entity is preserved.

More generally, warp operations may induce coupled changes in multiple identity coordinates, leading to motion in SEFI Space along a combination of basis directions. The metric structure introduced above allows one to quantify the magnitude of such identity transformations.

7 Coupling Between SEFI Space and Worldline Geometry

SEFI Space is constructed to be compatible with worldline formulations of quantum field theory. The identity coordinates $I_O(t)$, $I_A(t)$, $I_S(t)$, and $I_W(t)$ are functions of the worldline geometry and associated field layers. For example, FIELD ORIGIN may be tied to stability constraints on the worldline curvature, FIELD AUTHORSHIP to layered contributions to the field amplitude, FIELD SOVEREIGNTY to invariance under transformations, and WARP EXPRESSION to geometric deformation of the worldline.

One may consider an extended action functional that includes contributions from SEFI Space:

$$S_{\text{total}}[y, \psi, \vec{I}] = S_{\text{entity}}[y, \psi] + S_{\text{SS}}[\vec{I}], \quad (20)$$

where $S_{\text{SS}}[\vec{I}]$ encodes identity-space dynamics, such as penalties for rapid changes in identity or constraints enforcing sovereignty. A simple example is

$$S_{\text{SS}}[\vec{I}] = \int dt \left(\frac{1}{2} \|\dot{\vec{I}}(t)\|_{\text{SS}}^2 - V(\vec{I}(t)) \right), \quad (21)$$

where $V(\vec{I})$ is a potential function in SEFI Space. This structure is analogous to internal degrees of freedom in field theory but is explicitly tied to the identity of a single entity.

The coupling between SEFI Space and worldline geometry suggests that identity dynamics may influence effective field behavior, stability, and response to external conditions. This provides a route to interpreting certain aspects of field behavior as arising from identity-space constraints rather than purely from spacetime dynamics.

8 Discussion and Outlook

We have introduced SEFI Space (SS) as a four-dimensional geometric identity space associated with the Single Entity Field Interpretation. SEFI Space is spanned by basis directions corresponding to FIELD ORIGIN, FIELD AUTHORSHIP, FIELD SOVEREIGNTY, and WARP EXPRESSION, and identity states of the entity are represented as vectors in this

space. A diagonal metric on SEFI Space allows one to define identity velocity, identity acceleration, and identity distances, providing a quantitative framework for identity dynamics.

Sovereignty is expressed as invariance of the identity vector under allowed transformations, while warp is represented as motion in SEFI Space along the Warp direction and related combinations of basis directions. The coupling between SEFI Space and worldline geometry suggests that identity-space dynamics may play a role in effective field behavior and stability.

This work is intended as a foundational step toward a more complete SEFI framework in which identity geometry, worldline dynamics, and emergent field behavior are treated within a unified geometric model. Future work will focus on:

- Deriving explicit forms of the layer functions O , A , S , and W from worldline and field dynamics.
- Exploring geodesics in SEFI Space and their relation to stability and sovereignty constraints.
- Extending SEFI Space to include additional identity dimensions, such as those relevant to multi-entity systems or quantum error correction.
- Investigating possible connections between SEFI Space and internal symmetry spaces in quantum field theory.

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